

Expect the unexpected: probabilistic failure of WLP in monomial algebras

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Some historical context: WLP in the early days of combinatorial algebra

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Let $R = \mathbb{C}[x_1, \dots, x_n]$, $I = (x_1^{a_1}, \dots, x_n^{a_n})$, $L = x_1 + \cdots + x_n$ and $A = R/I$. Then the multiplication maps

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If the statement holds for an ideal I , $j = 1$ and a general linear form L , we say A has the weak Lefschetz property (WLP)

Some historical context: everything should have the WLP!

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- ⑥ (Migliore–Miró-Roig–Nagel, 2011) For every n , either $A(n)$ does not exist, **or $A(n) = n$** .

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In the same paper, they showed that $L = x_1 + \cdots + x_n$ is enough to check the WLP/SLP of monomial algebras

The role of algebra, geometry and combinatorics in the study of WLP

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If the motto in **geometry** is that “algebras fail the WLP **for a reason**”, then what is the **combinatorial** meaning of **failure** of WLP?

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spoilers: 1) most likely yes, 2) an example is an analogue of the bipartite property

Question 1: maybe we shouldn't expect failure

During the Lefschetz meeting in 2023, Nagel asked the following question:

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Given a probability space $\mathcal{P}$ of monomial algebras, what is the probability that an algebra fails the WLP?

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“Although the WLP appears to be present with high probability most of the time, we propose in Conjecture 3.9 that there is one scenario where the produced algebras fail to have the WLP with high probability” (Nagel–Petrović, 2024)

Question 1: a combinatorial perspective

A simplicial complex Δ is a collection of subsets of $[n]$ closed under inclusion.

$$I_{\Delta} = \left(\prod_{i \in \sigma} x_i : \sigma \notin \Delta \right) \subseteq \mathbb{K}[x_1, \dots, x_n]$$

Δ can be seen as a topological space, and in particular it has homology groups. The **face numbers** f_i of Δ are the number of sets of size $i + 1$ in Δ

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Theorem (H, 2026)

Let Δ be a simplicial complex of dimension $d > 0$. If $H_d(\Delta; \mathbb{K}) \neq 0$ and $f_{d-1} > f_d$, then the algebra

$$A = \frac{\mathbb{K}[x_1, \dots, x_n]}{I_{\Delta} + (x_1^{d+2}, \dots, x_n^{d+2})} \quad \text{fails the WLP}$$

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When Δ is the boundary of a simplex, we recover the original statement that **monomial almost complete intersections fail the WLP**

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Theorem (H, 2026)

For every d , there exists a real number $c_d < d + 1$ such that if $p = \frac{c}{n}$ in the model above for some $c_d < c < d + 1$, then

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\Delta: \frac{\mathbb{K}[x_1, \dots, x_n]}{I_\Delta + (x_1^{d+2}, \dots, x_n^{d+2})} \text{ fails the WLP} \right) = 1$$

The proof is showing the assumptions on the previous theorem are **very mild in a very concrete sense**

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For $d \geq 2$, the number c_d is computed by finding the unique positive solution to the equation

$$(d+1)(x+1)e^{-x} + x(1-e^{-x})^{d+1} = d+1$$

The size of the (nonempty!) interval $(c_d, d+1)$ converges to 0 very quickly

d	1	2	3	4	5	6	7	8
c_d	1	2.783	3.91	4.962	5.984	6.993	7.997	8.998

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Conjecture: "Everything fails!"

Let $t(d, n)$ be the number of artinian monomial algebras A of codimension n such that $x_1^d = \cdots = x_n^d = 0$ in A , and $w(d, n)$ be the subset of these algebras failing the WLP. Then for every $n \geq 3$ and every $d \geq 2$

$$\lim_{n \rightarrow \infty} \frac{w(d, n)}{t(d, n)} = 1 \quad \text{and} \quad \lim_{d \rightarrow \infty} \frac{w(d, n)}{t(d, n)} = 1.$$

Question 2: As always, its all about inverse systems

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Inverse systems

Let $R = \mathbb{K}[x_1, \dots, x_n]$, $S = \mathbb{K}[y_1, \dots, y_n]$ and assume R acts on S via contraction, which is defined linearly from the definition on monomials:

$$x_1^{a_1} \dots x_n^{a_n} \circ y_1^{b_1} \dots y_n^{b_n} = \begin{cases} y_1^{b_1-a_1} \dots y_n^{b_n-a_n} & \text{if } a_i \leq b_i \text{ for all } i \\ 0 & \text{otherwise} \end{cases}$$

For an R -module $M \subseteq S$, the inverse system of M is the set

$$\text{Ann}(M) = \{f \in R \quad : \quad f \circ m = 0 \quad \text{for all } m \in M\}.$$

- ① M is finite if and only if $I = \text{Ann}(M)$ is artinian
- ② M is cyclic if and only if $I = \text{Ann}(M)$ is artinian Gorenstein. A generator F of M is then called the Macaulay dual of I

Question 2: An unexpected intuition

Definition (Cook–Harbourne–Migliore–Nagel, 2018)

Let $X \subseteq \mathbb{P}^n$ be a finite set of points. We say X admits an **unexpected** hypersurface of degree d with a general point P of multiplicity m if

$$\dim(I_X \cap I_P^m)_d > \max\left(0, \dim(I_X)_d - \binom{m+n-1}{n}\right)$$

Theorem (Cook–Harbourne–Migliore–Nagel, 2018)

A set of points in $\mathbb{P}^2$ admits an unexpected curve if and only if a certain algebra fails the SLP in a specific degree.

The intuition for this equivalence is to look at inverse systems and a result of Emsalem-Iarrobino

Question 2: What is the transpose of multiplication?

Lemma (H, 2026)

Let $L = x_1 + \cdots + x_n$ and $A = R/I$, where I is a monomial ideal. Then, after possibly deleting rows and columns, the matrix representing the multiplication map

$$\times L^\top : A_i \rightarrow A_{i-1}$$

is the same as the matrix representing the map

$$\circ L : S_{-i} \rightarrow S_{-i+1}$$

We can now use inverse systems as a way to try to find weird phenomena that explains failure of WLP! An element in the kernel of $\times L^\top$ is an element f such that $I + L \subseteq \text{Ann}(f)$

Question 2: a concept of unexpectedness for simplicial complexes

Definition

Let I_Δ be a square-free Gorenstein* monomial ideal. A sequence of elements $f_1, \dots, f_s$ is an **a-unexpected system of parameters** of Δ if

- 1 $f_1 = L$
- 2 $f_1, \dots, f_s$ is a system of parameters of I_Δ
- 3 $x_i^a \in I_\Delta + (f_1, \dots, f_s)$ for every i

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Theorem (H, 2026)

If Δ admits an a -unexpected sop, then the algebra

$$\frac{R}{I_\Delta + (x_1^a, \dots, x_n^a)}$$

fails the WLP.

Question 2: Failing for a reason

Assume Δ is a triangulation of a d -dimensional sphere. We say Δ is **balanced** if we can color the graph of Δ using $d + 1$ colors.

Theorem (H, 2026)

- 1 Δ admits a 2-unexpected sop if and only if Δ is balanced.
- 2 the only 2-unexpected sops that Δ admits arise from the colorings of the graph of Δ using $d + 1$ colors

The octahedron

Let $I_\Delta = (x_1x_2, x_3x_4, x_5x_6)$, $f_1 = L$, $f_2 = x_1 + x_2$, $f_3 = x_3 + x_4$, $f_4 = x_5 + x_6$.

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- 3 $\text{Ann}(F) = I_\Delta + (f_1, \dots, f_4)$

Check out the papers

